

Name:

Class:

Date:

GCSE TOPIC TEST

PHYSICS

H

Higher Tier

4.4 Atomic Structure

Materials

For this paper you must have:

- a ruler
- a scientific calculator
- the Physics Equation sheet

Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- In all calculations, show clearly how you work out your answer.

Information

- The maximum mark for this paper is 40.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

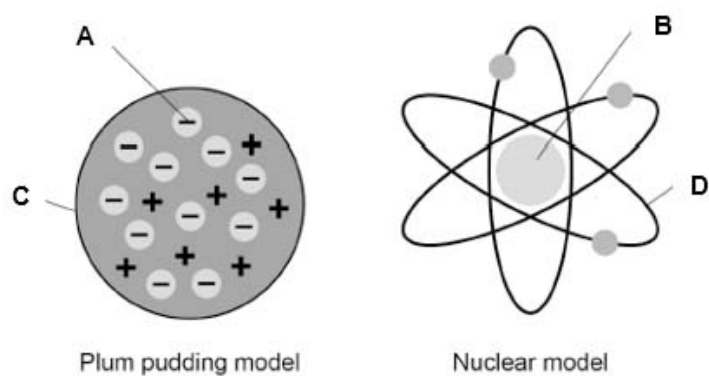
01

Explain the differences between the properties of alpha, beta and gamma radiations.

[6 marks]

Figure 1 shows two models of the atom.

Figure 1



Tick **one** box to identify each label on the diagram.

A

- ☐ atom
- ☐ electron
- ☐ nucleus
- ☐ neutron
- ☐ orbit
- ☐ proton

B

- ☐ atom
- ☐ electron
- ☐ nucleus
- ☐ neutron
- ☐ orbit
- ☐ proton

C

- ☐ atom
- ☐ electron
- ☐ nucleus
- ☐ neutron
- ☐ orbit
- ☐ proton

D

- ☐ atom
- ☐ electron
- ☐ nucleus
- ☐ neutron
- ☐ orbit
- ☐ proton

[4 marks]

03

A householder is worried about the radiation emitted by the granite worktop in his kitchen.

1 kg of granite has an activity of 1250 Bq. The kitchen worktop has a mass of 180 kg.

(a) Calculate the activity of the kitchen worktop in Bq.

Activity = _____ Bq

(2)

The average total radiation dose per year in the UK is 2.0 millisieverts.

Table 1 shows the effects of radiation dose on the human body.

Table 1

Radiation dose in millisieverts	Effects
10 000	Immediate illness; death within a few weeks
1000	Radiation sickness; unlikely to cause death
100	Lowest dose with evidence of causing cancer

The average radiation dose from the granite worktop is 0.003 millisieverts per day.

- (b)** Explain why the householder should **not** be concerned about his yearly radiation dose from the granite worktop.

One year is 365 days.

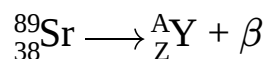
(2)

[4 marks]

04

A strontium-89 nucleus (Sr) emits a beta particle (β) and turns into an yttrium nucleus (Y).

This can be represented by the equation:



What are the values of A and Z in the equation?

A = _____

Z = _____

[2 marks]

05

A teacher used a Geiger-Muller tube and counter to measure the number of counts in 60 seconds for a radioactive rock.

The counter recorded 819 counts in 60 seconds. The background radiation count rate was 0.30 counts per second.

Calculate the count rate for the rock.

Count rate = per second

[3 marks]

06

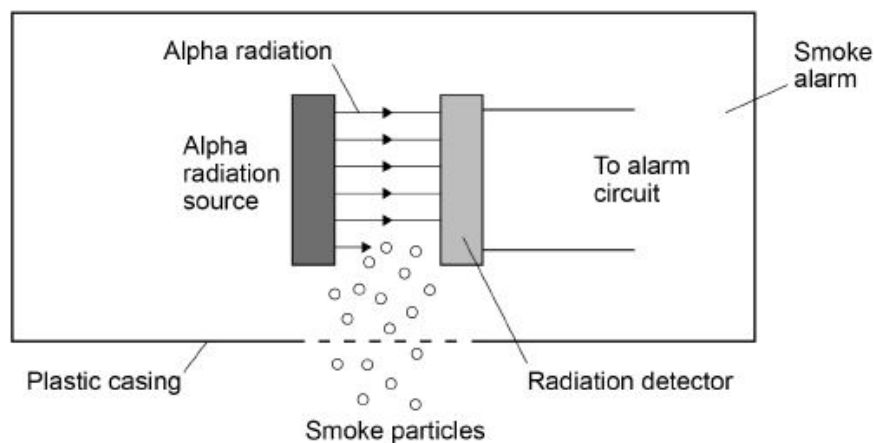
Explain why the total positive charge in every atom of an element is always the same.

[2 marks]

07.1

Smoke alarms contain an alpha radiation source and a radiation detector. **Figure 2** shows part of the inside of a smoke alarm.

Figure 2



The smoke alarm stays off while alpha radiation reaches the detector.

Why does the alarm switch on when smoke particles enter the plastic casing?

[1 mark]

07.2

Why is it safe to use a source of alpha radiation in a house?

[1 mark]

07.3

The smoke alarm would not work with a radiation source that emits beta or gamma radiation.

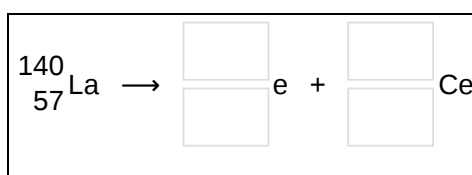
Explain why.

[2 marks]

08

Lanthanum-140 can emit beta radiation and change into cerium.

Complete the equation showing the decay of lanthanum (La) 140 into cerium (Ce).



[2 marks]

09

Polonium-210 ($^{210}_{84}\text{Po}$) is a radioactive isotope that decays by emitting alpha radiation.

A sample of polonium-210 was left for 414 days.

After this time it had a mass of 1.45×10^{-4} g

The half-life of polonium-210 is 138 days.

Calculate the initial mass of the sample.

Initial mass = g

[3 marks]

Figure 3 shows a patient who has been injected with a radioactive source for medical diagnosis.

Figure 3



Explain the ideal properties of a radioactive source for use in medical diagnosis.

[4 marks]

11.1

Nuclear power stations use the energy released by nuclear fission to generate electricity.

Nuclear fission releases energy.

Describe the process of nuclear fission inside a nuclear reactor.

[4 marks]

11.2

A new type of power station is being developed that will generate electricity using nuclear fusion.

Explain how the process of nuclear fusion leads to the release of energy.

[2 marks]